

CDP & LLDP

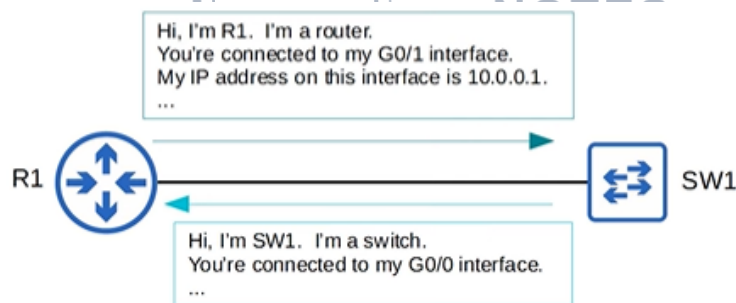
- Topics covered:
 - Intro Layer 2 Discovery Protocols
 - Cisco Discovery Protocols (CDP)
 - Link Layer Discovery Protocols (LLDP)
 - Wireshark Captures

Layer 2 Discovery Protocols

Introduction to Layer 2 Discovery Protocols

Layer 2 discovery protocols are used to share and learn information about directly connected devices.

[R1] <----- Layer 2 frames -----> [SW1]



R1 sends:

- Hostname
- Device type
- Interface ID
- IP address

SW1 sends:

- Hostname
- Device type
- Interface ID

(no IP address on switch interface)

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These protocols operate at Layer 2, meaning:

- They do not use IP packets.
- Information is sent inside Layer 2 frames.

However, they can still share Layer 3 information like IP addresses.

Two main protocols:

1. **CDP** → Cisco proprietary
2. **LLDP** → Industry standard (IEEE 802.1AB)

If the network has mixed vendors, LLDP must be used.

Because these protocols share device information, they can be considered a security risk and are sometimes disabled.



Cisco Discovery Protocols (CDP)

CDP Overview

CDP (Cisco Discovery Protocol):

- Cisco proprietary
- Enabled **by default** on Cisco devices
- Runs on routers, switches, firewalls, IP phones

Key Behaviour

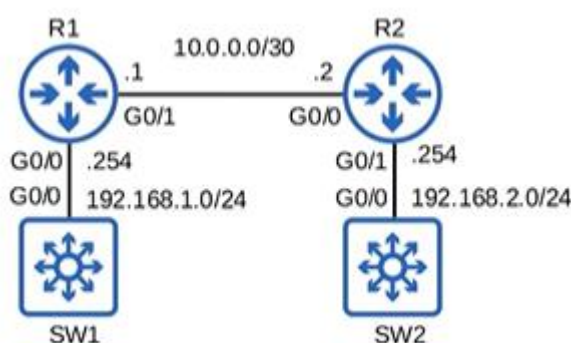
- CDP messages are sent to multicast MAC: 0100.0CCC.CCCC
- When a device receives a CDP message, it processes and discards the message. It does NOT forward it to other devices.
- Only directly connected neighbours form CDP relationships

CDP Timers

- By Default, CDP messages are sent once every 60 seconds
- By Default, the CDP hold time is 180 seconds
- If no message is received for 180 seconds:
Neighbour is removed from table

Default version: CDP version 2

CDP Demonstration Topology



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CDP Show Commands

A] show cdp

```
R1#show cdp
Global CDP information:
  Sending CDP packets every 60 seconds
  Sending a holdtime value of 180 seconds
  Sending CDPv2 advertisements is enabled
```

Explanation

This command shows-

- CDP timer (60 seconds)
- Hold time (180 seconds)
- CDP version (v2 enabled)

B] show cdp traffic

```
R1#show cdp traffic
CDP counters :
  Total packets output: 105, Input: 112
  Hdr syntax: 0, Chksum error: 0, Encaps failed: 0
  No memory: 0, Invalid packet: 0,
  CDP version 1 advertisements output: 0, Input: 0
  CDP version 2 advertisements output: 105, Input: 112
R1#
```

Explanation:

Shows how many CDP messages-

- Sent (output)
- Received (input)

All messages shown are CDP version 2.

C] show cdp interface

```
R1#show cdp interface
GigabitEthernet0/0 is up, line protocol is up
  Encapsulation ARPA
  Sending CDP packets every 60 seconds
  Holdtime is 180 seconds
GigabitEthernet0/1 is up, line protocol is up
  Encapsulation ARPA
  Sending CDP packets every 60 seconds
  Holdtime is 180 seconds
GigabitEthernet0/2 is administratively down, line protocol is down
  Encapsulation ARPA
  Sending CDP packets every 60 seconds
  Holdtime is 180 seconds
GigabitEthernet0/3 is administratively down, line protocol is down
  Encapsulation ARPA
  Sending CDP packets every 60 seconds
  Holdtime is 180 seconds

cdp enabled interfaces : 4
interfaces up          : 2
interfaces down        : 2
```

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Explanation:

This command shows per-interface CDP status-

- Which interfaces have CDP enabled
- Timer and holdtime per interface
- Interface state (up/down)

“Encapsulation ARPA” refers to Ethernet encapsulation (Ethernet II).

CDP Neighbour Table

A] show cdp neighbors

```
R1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
                  D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID         Local Intrfce   Holdtme    Capability Platform Port ID
SW1                Gig 0/0         153        R S I       Gig 0/0
R2                 Gig 0/1         146        R B         Gig 0/0

Total cdp entries displayed : 2
R1#
```

Explanation:

Each column provides specific information-

- Device ID → Neighbour hostname (SW1, R2)
- Local Interface → Interface on R1
- Holdtime → Countdown timer (resets when message received)
- Capability:
 - R → Router
 - S → Switch
- Platform → Device model (not shown in virtual devices)
- Port ID → Neighbour's interface

Key Understanding:

- SW1 is connected to **R1 G0/0**
- R2 is connected to **R1 G0/1**

This command helps identify physical connections.

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Platform Example (from Packet Tracer)

```
R1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID        Local Intrfce  Holdtme  Capability  Platform  Port ID
R2               Gig 0/1       124      R           C2900     Gig 0/0
SW1              Gig 0/0       126      S           2960      Gig 0/1
```

Explanation:

Here, platform shows actual device models:

- Router → C2900
- Switch → 2960

B] show cdp neighbors detail

Detailed Neighbour Information

```
R1#show cdp neighbors detail
-----
Device ID: SW1
Entry address(es):
Platform: Cisco, Capabilities: Router Switch IGMP
Interface: GigabitEthernet0/0, Port ID (outgoing port): GigabitEthernet0/0
Holdtime : 174 sec

Version :
Cisco IOS Software, vlos_l2 Software (vlos_l2-ADVENTERPRISEK9-M), Version 15.2(4.0.55)E, TEST ENGINEERING
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2015 by Cisco Systems, Inc.
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advertisement version: 2
VTP Management Domain: ''
Native VLAN: 1
Duplex: full

-----
Device ID: R2
Entry address(es):
  IP address: 10.0.0.2
Platform: Cisco, Capabilities: Router Source-Route-Bridge
Interface: GigabitEthernet0/1, Port ID (outgoing port): GigabitEthernet0/0
Holdtime : 163 sec

Version :
Cisco IOS Software, IOSv Software (VIOS-ADVENTERPRISEK9-M), Version 15.6(2)T, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
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advertisement version: 2
Duplex: full
Management address(es):
  IP address: 10.0.0.2

Total cdp entries displayed : 2
```

Explanation:

This command provides more detailed information-

For SW1:

- IOS version
- Native VLAN
- Duplex setting

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For R2:

- IP address (10.0.0.2)

CDP can detect:

- Native VLAN mismatches
- Duplex mismatches

C] Show cdp entry <name>

```
R1#show cdp entry R2
-----
Device ID: R2
Entry address(es):
  IP address: 10.0.0.2
Platform: Cisco , Capabilities: Router Source-Route-Bridge
Interface: GigabitEthernet0/1, Port ID (outgoing port): GigabitEthernet0/0
Holdtime : 178 sec

Version :
Cisco IOS Software, IOSv Software (VIOS-ADVENTERPRISEK9-M), Version 15.6(2)T, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
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advertisement version: 2
Duplex: full
Management address(es):
  IP address: 10.0.0.2
```

Explanation:

- Same information as show cdp neighbours detail
- But only for a single specified neighbour

CDP Commands Summary

- R1# **show cdp**
→ shows basic information about CDP (timers, version)
- R1# **show cdp traffic**
→ displays how many CDP messages have been sent and received
- R1# **show cdp interface**
→ displays which interfaces CDP is enabled on
- R1# **show cdp neighbors**
→ lists CDP neighbors and some basic information about each neighbor
- R1# **show cdp neighbors detail**
→ lists each CDP neighbor with more detailed information
- R1# **show cdp entry name**
→ displays the same info as above, but for the specified neighbor only

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CDP Configuration

CDP is enabled by default, and also enabled on each interface by default.

Global Configuration

- Enable CDP: `cdp run`
- Disable CDP: `no cdp run`

Interface Configuration

- Enable CDP on interface: `cdp enable`
- Disable CDP on interface: `no cdp enable`

Timer Configuration

- Change message interval: `cdp timer <seconds>`
- Change hold time: `cdp holdtime <seconds>`

CDP Version

- Enable version 2 (default): `cdp advertise-v2`
- Disable (use version 1): `no cdp advertise-v2`



Link Layer Discovery Protocol (LLDP)

LLDP Overview

LLDP is an industry-standard Layer 2 discovery protocol (IEEE 802.1AB).

- CDP was created first (Cisco proprietary).
- LLDP was later developed as a vendor-neutral alternative.
- Used in multi-vendor networks.

A device can run:

- CDP and LLDP at the same time

However:

- LLDP is disabled by default on Cisco devices
- Must be manually enabled

Basic Operation

LLDP works like CDP:

- Devices periodically send frames to neighbours
- Frames contain:
 - Hostname
 - Device type
 - Interface ID
 - Other details

Multicast MAC Address: 0180.C200.000E

- LLDP messages are sent to this multicast MAC
- When received:
 - Device processes and discards
 - Does not forward

Only directly connected devices become neighbours

LLDP Timers

- By Default, LLDP message are sent once every 30 seconds
- By Default, the LLDP Hold time 120 seconds
- Reinitialization delay → 2 seconds

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Reinitialization Delay (Important context)

- When LLDP is enabled:
 - It does not start immediately
 - Waits for 2 seconds
- Helps in situations where LLDP is rapidly enabled/disabled

LLDP Configuration

LLDP must be enabled in two steps.

Step 1 – Global Enable

Enables LLDP globally:

```
R1(config)# lldp run
```

To disable:

```
R1(config)# no lldp run
```

Step 2 – Interface Enable

LLDP uses separate commands for send and receive.

```
R1(config-if)# lldp transmit
```

```
R1(config-if)# lldp receive
```

Explanation:

- lldp transmit → interface sends LLDP frames
- lldp receive → interface processes incoming LLDP frames

Unlike CDP:

- CDP uses **one command**
- LLDP requires **both commands**

Step 3 – Timer Command

Configure the LLDP timer: `R1(config)# lldp timer seconds`

Configure the LLDP holdtime: `R1(config)# lldp holdtime seconds`

Configure the LLDP reinit timer: `R1(config)# lldp reinit seconds`

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LLDP Show Commands

A] show lldp

```
R1#show lldp

Global LLDP Information:
  Status: ACTIVE
  LLDP advertisements are sent every 30 seconds
  LLDP hold time advertised is 120 seconds
  LLDP interface reinitialisation delay is 2 seconds
```

Explanation

This confirms:

- LLDP is enabled
- Timer = 30 sec
- Holdtime = 120 sec
- Reinit delay = 2 sec

B] show lldp traffic

```
R1#show lldp traffic

LLDP traffic statistics:
  Total frames out: 4
  Total entries aged: 0
  Total frames in: 3
  Total frames received in error: 0
  Total frames discarded: 0
  Total TLVs discarded: 0
  Total TLVs unrecognized: 0
R1#
```

Explanation

Shows LLDP message activity:

- Frames sent (out)
- Frames received (in)
- Errors or discarded frames

Here:

- Very few frames because LLDP was just enabled

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C] show lldp interface

```
R1#  
R1#show lldp interface  
  
GigabitEthernet0/0:  
Tx: enabled  
Rx: enabled  
Tx state: IDLE  
Rx state: WAIT FOR FRAME  
  
GigabitEthernet0/1:  
Tx: enabled  
Rx: enabled  
Tx state: IDLE  
Rx state: WAIT FOR FRAME  
  
GigabitEthernet0/2:  
Tx: enabled  
Rx: enabled  
Tx state: INIT  
Rx state: WAIT PORT OPER  
  
GigabitEthernet0/3:  
Tx: enabled  
Rx: enabled  
Tx state: INIT  
Rx state: WAIT PORT OPER
```

Explanation

For each interface:

- Tx / Rx → whether LLDP is enabled
- Tx state → waiting to send next frame
- Rx state → waiting to receive frame

This confirms LLDP is active on interfaces.

D] show lldp neighbors

```
R1#show lldp neighbors  
Capability codes:  
  (R) Router, (B) Bridge, (T) Telephone, (C) DOCSIS Cable Device  
  (W) WLAN Access Point, (P) Repeater, (S) Station, (O) Other  
  
Device ID      Local Intf    Hold-time    Capability    Port ID  
SW1            Gi0/0         120          B             Gi0/0  
R2             Gi0/1         120          R             Gi0/0  
  
Total entries displayed: 2
```

Explanation (Field-by-Field)

- Device ID → neighbor hostname
- Local Interface → interface on R1
- Hold-time → shows configured value (120), not counting down
- Capability: R → Router & B → Bridge (switch)

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Important Detail

- Switch uses **B (Bridge)** instead of “Switch”
- In this output, SW1 capability is not shown properly (environment limitation)

E] show lldp neighbors detail

```

R1#show lldp neighbors detail
-----
Local Intf: Gi0/0
Chassis id: 0c04.41d2.1a00
Port id: Gi0/0
Port Description: GigabitEthernet0/0
System Name: SW1

System Description:
Cisco IOS Software, vios_12 Software (vios_12-ADVENTERPRISEK9-M), Version 15.2(4.0.55)E, TEST ENGINEERIN
Technical Support: http://www.cisco.com/techsupport
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Time remaining: 99 seconds
System Capabilities: B,R
Enabled Capabilities - not advertised
Management Addresses - not advertised
Auto Negotiation - not supported
Physical media capabilities - not advertised
Media Attachment Unit type - not advertised
Vlan ID: - not advertised

-----
Local Intf: Gi0/1
Chassis id: 0c04.418d.a400
Port id: Gi0/0
Port Description: GigabitEthernet0/0
System Name: R2

System Description:
Cisco IOS Software, IOSv Software (VIOS-ADVENTERPRISEK9-M), Version 15.6(2)T, RELEASE SOFTWARE (fc2)
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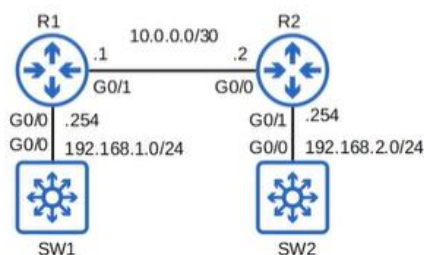
Time remaining: 92 seconds
System Capabilities: B,R
Enabled Capabilities: R
Management Addresses:

```

Explanation

This command shows additional details per neighbor:

- System name (hostname)
- OS version
- Interface description
- Time remaining (actual countdown)



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Key Difference from CDP

LLDP shows two capability fields:

- System Capabilities
 - What the device *can do*
 - Example: B (bridge), R (router)
- Enabled Capabilities
 - What is currently active
- In this case: “not advertised” appears due to environment behaviour

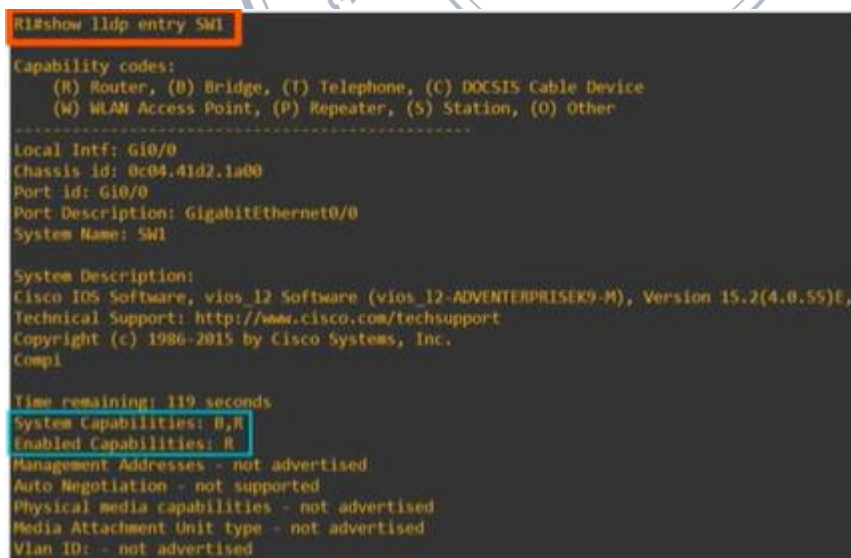
Important Note from Explanation

- SW1 is a multilayer switch
 - So it supports both:
 - Bridge (switch)
 - Router
- If routing is enabled:
 - Router capability appears under enabled capabilities

CDP vs LLDP Difference Here

- CDP → shows VTP info
- LLDP → does NOT show VTP
 - Because VTP is Cisco proprietary

F] show lldp entry <name>



```
R1#show lldp entry SW1
Capability codes:
  (R) Router, (B) Bridge, (T) Telephone, (C) DOCSIS Cable Device
  (W) WLAN Access Point, (P) Repeater, (S) Station, (O) Other
-----
Local Intf: Gi0/0
Chassis id: 0c04.41d2.1a00
Port id: Gi0/0
Port Description: GigabitEthernet0/0
System Name: SW1

System Description:
Cisco IOS Software, vios_12 Software (vios_12-ADVENTERPRISEK9-M), Version 15.2(4.0.55)E,
Technical Support: http://www.cisco.com/techsupport
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Compl

Time remaining: 119 seconds
System Capabilities: B,R
Enabled Capabilities: R
Management Addresses - not advertised
Auto Negotiation - not supported
Physical media capabilities - not advertised
Media Attachment Unit type - not advertised
Vlan ID: - not advertised
```

Explanation

- Same output as: show lldp neighbors detail
- But only for one specific neighbor

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LLDP Commands Summary

- **R1# show lldp**
→ shows basic information about LLDP (timers, version)
- **R1# show lldp traffic**
→ displays how many LLDP messages have been sent and received
- **R1# show lldp interface**
→ displays which interfaces LLDP tx/rx is enabled on
- **R1# show lldp neighbors**
→ lists LLDP neighbors and some basic information about each neighbor
- **R1# show lldp neighbors detail**
→ lists each LLDP neighbor with more detailed information
- **R1# show lldp entry name**
→ displays the same info as above, but for the specified neighbor only



Wireshark Captures

CDP Frame

```
> Frame 12: 369 bytes on wire (2952 bits), 369 bytes captured (2952 bits) on interface -, id 0
▼ IEEE 802.3 Ethernet
  > Destination: CDP/VTP/DTP/PagP/UDLD (01:00:0c:cc:cc:cc)
  > Source: 0c:04:41:47:57:00 (0c:04:41:47:57:00)
  Length: 355
  > Logical-Link Control
  ▼ Cisco Discovery Protocol
    Version: 2
    TTL: 180 seconds
    Checksum: 0xee0f [correct]
    [Checksum Status: Good]
    > Device ID: R1
    > Software Version
    > Platform: Cisco
    > Addresses
    > Port ID: GigabitEthernet0/0
    ▼ Capabilities
      Type: Capabilities (0x0004)
      Length: 8
      ▼ Capabilities: 0x00000005
        ....1 = Router: Yes
        ...0. = Transparent Bridge: No
        ...1.. = Source Route Bridge: Yes
        ...0... = Switch: No
        ...0.... = Host: No
        ...0..... = IGMP capable: No
        ...0..... = Repeater: No
        ...0..... = VoIP Phone: No
        ...0..... = Remotely Managed Device: No
        ...0..... = CVTA/STP Dispute Resolution/Cisco VT Camera: No
        ...0..... = Two Port Mac Relay: No
    > IP Prefixes: 1
    > Duplex: Full
    > Management Addresses
```

This is a CDP messages sent from R1 to SW1.

Destination MAC: 0100.0CCC.CCCC

Protocol: CDP v2

TTL: 180 seconds

Fields:

- Device ID
- Platform
- IP address
- Port ID
- Capabilities

Explanation

- TTL = CDP holdtime
- Capabilities show device roles
- No IP packet inside (Layer 2 only)

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LLDP Frame

```
> Frame 466: 325 bytes on wire (2600 bits), 325 bytes captured (2600 bits) on interface -, id 0
> Ethernet II, Src: 0c:04:41:d2:1a:00 (0c:04:41:d2:1a:00), Dst: LLDP_Multicast (01:80:c2:00:00:0e)
  > Link Layer Discovery Protocol
    > Chassis Subtype = MAC address, Id: 0c:04:41:d2:1a:00
    > Port Subtype = Interface name, Id: Gi0/0
    > Time To Live = 120 sec
    > System Name = SW1
    > [truncated]System Description = Cisco IOS Software, vios_12 Software (vios_12-ADVENTERPRISEK9-M), Versic
    > Port Description = GigabitEthernet0/0
    > Capabilities
      0000 111. .... = TLV Type: System Capabilities (7)
      .... 0000 0100 = TLV Length: 4
      > Capabilities: 0x0014
        .... 0 = Other: Not capable
        .... 0 = Repeater: Not capable
        .... 1 = Bridge: Capable
        .... 0 = WLAN access point: Not capable
        .... 1 = Router: Capable
        .... 0 = Telephone: Not capable
        .... 0 = DOCSIS cable device: Not capable
        .... 0 = Station only: Not capable
      > Enabled Capabilities: 0x0010
        .... 0 = Other: Not capable
        .... 0 = Repeater: Not capable
        .... 0 = Bridge: Not capable
        .... 0 = WLAN access point: Not capable
        .... 1 = Router: Capable
        .... 0 = Telephone: Not capable
        .... 0 = DOCSIS cable device: Not capable
        .... 0 = Station only: Not capable
    > End of LLDPDU
```

Destination MAC: 0180.C200.000E

Protocol: LLDP

Fields:

- System Name (SW1)
- System Description
- Port ID
- TTL: 120 seconds
- Capabilities

Explanation

- TTL = LLDP holdtime
- Shows hostname and capabilities
- No IP packet inside